



COMSATS University Islamabad, Lahore Campus
Department of Computer Sciences

Project – Semester Spring 2025

Course Title	Software Testing			Course Code	CSE455	Credit Hours:	3(2,1)
Course Instructor	Ms. Yella Mehroze			Program Name	BSE		
Semester	8 th	Batch:	FA21-BSE	Section	A ,B	Type	Lab
Total Marks	15	Obtained Marks:		Date			
Name				Registration ID			

Question1:

[Marks 15]

[CLO <4>]; Implement a program using basic programming constructs

Bloom Taxonomy Level<Applying>

Statement :

Make use of Jira to apply the following concepts and construct the artifacts in Jira and Zephyr scale for test case management.

1. Scenario Creation

- Define a realistic software project scenario (do not write a scenario from your final year project as it may lead to similarity between students)
- Briefly describe the key features (enlisting requirements), users, and objectives of the project (not more than 10 lines).
- *Add this scenario in the word document you will submit.*

2. Jira Artifacts

I-Epic

- Write one Epic representing a major feature or goal.

II-User Story

- Break down the Epic into at 1 user stories.
- Example: *"As a customer, I want to filter books by genre so that I can narrow down my search."*

II-Issue (Task/Bug)

- Create one Task (e.g., "Implement book search functionality").
- Create one Bug (e.g., "Search results not updating when filters are applied").
- Add one more issue type from your own and name it by your name.
- *Insert a screenshot of all artifacts in the word document you will submit.*

3. Jira Project Setup

I-Sprint Planning

- Sprint Duration: 2 weeks (set Start Date and End Date).
- WIP (Work in Progress) Limit: Set a limit (e.g., "Max 3 In-Progress tasks per developer").
- Do not end the sprint before Mid Exam (End date should not be before exam).

II-Workflow Diagram

- Design a custom workflow (e.g., To Do → In Progress → Code Review → Testing → Done). Add headings as per your scenario.
- *Add a screenshot of your workflow from Jira in the Word document that you will submit.*
- Draw a proper workflow diagram against your scenario. This diagram must exhibit that is drawn against your particular scenario.
- *Add a screenshot of the workflow diagram from Jira in the Word document that you will submit.*

III-Automation Rule

- Create one automation rule (e.g., "When a Bug is moved to *In Progress*, assign it to the abc@gmail.com").
- *Provide a one or two liner description of your rule and screenshot of the rule configuration in your word document.*

4. Confluence Documentation

- Create a Confluence page documenting:
- Document Owner (Your Name)
- Project overview (scenario, objectives).
- Attach Links to Jira Epic and Bug.
- Show your jira timeline in confluence (Hint : Create->Page->'More Elements'->'Jira Timeline')
- Add the date of your MID TERM LAB EXAM in Confluence using more elements option
- *Export your confluence and add in the word document.*

5. Jira Dashboard

Dashboard Gadgets

1. Pie Chart (Issue types: Story/Bug/Task/Your self added issue ('that issue was on your name')).
2. Sprint Burndown Chart.
3. Zephyr Test Execution Summary (any from gadgets of your choice).
4. Days Remaining in Sprint Gadget
5. One more gadget of your choice.
6. *Add a screenshot of your gadget in word document.*

6. GUI Mockup

- Attach a mockup (using Figma, Balsamiq, or hand-drawn) against one feature or say story for your scenario. The mockup should contain web elements like , dropdown/toggle button/text field e.t.c so you may write a detailed test case against it.
- *Provide screenshot of mockup in your word document.*

7. Zephyr Scale Test Management

I-Test Case Design

- Write two detailed test cases covering functional requirements against your scenario (story e.t.c).
- Step by step methodology should be followed.
- Design test cases such that you have test data for test case steps.

II- Test Cycle Creation & Execution

- Create a Test Cycle (e.g., "Sprint 1 - Search Feature Testing").
- Assign test cases to the cycle.
- Execute tests and log results (Pass/Fail/Blocked). You must both fail and pass test cases (either for different steps within a test case or do it for different test cases)

- Add comments for failed tests (e.g., "Failed due to missing error handling").

III-Defect Reporting

- For failed tests, log a Jira Bug directly from Zephyr.
- Include the following.
 - Steps to Reproduce (from the test case).
 - Actual vs. Expected Result.
 - Severity (Critical/Major/Minor).

IV-Traceability & Reports

- Link test cases to the Jira user story/epic.
- Generate Zephyr Reports:
 - Test Execution Summary (Pass/Fail/Blocked percentages).
 - Defect Distribution (Bugs by severity).
 - Traceability reports
 Any other report of your choice.
- *Add a 'step by step' description screenshot of test case in word document.*
- *Add the screenshot of test coverage (that shows percentage of overall progress like : 80 percent complete e.t.c in your word document).*
- *Add a clip containing your video in which you are elaborating your test case management using zephyr scale and logging bug report against it. Add the drive link of this video in your word document. The clip must not exceed two minutes . Record your screen with your voice and video for this part.*

Instructions

- ✚ Copying project idea / scenario / report from one another is strictly not allowed. Write on your own/ Don't plagiarize the scenario from any source. Try to think of a unique scenario. For project, no late submissions will be entertained.
- ✚ Deadline for both sections : Sunday April 6,2025
- ✚ Google classroom will stop accepting submissions after the deadline.
- ✚ VIVA VOCE will be conducted for each student in the lab slots of April 7th and April 9th respectively for both sections.
- ✚ The project and the VIVA VOCE carry absolute weightage in Mid Lab Exam, as discussed in detail in class.